

Part II gate scoping in Lamport notation

Minimal logic, momentum-source obstruction, and executed route

Fresh reconstruction of `part_ii/gate_II11_scoping.tex`

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This module records the exact logical gate and the obstruction that dictates the thermal mean-shift method. The two transfer lemmas proposed by the scoping document are not left as promises: their complete proofs appear as Steps $\langle 1 \rangle 10$ and $\langle 1 \rangle 13$ of the V4 Lamport module.

Contents

1	The gate and what is logically sufficient	1
2	Why momentum cannot be inserted as a sharp-time Gaussian source	2
3	The executed thermal two-torus route	3
4	Scope conclusion	4

1 The gate and what is logically sufficient

For fixed coarse scale M , let α_M^N be the D4 lattice refinement and β_M the continuum embedding. The gate is

$$\lim_{N \rightarrow \infty} \omega_N(\alpha_M^N(W_M(\xi))) = \omega_{L,\lambda}^{\text{ct}}(\beta_M(W_M(\xi))) \quad (\xi \in h_{M,L}). \quad (\text{G})$$

$\langle 1 \rangle 1$ Identify the minimum statement that must be proved.

$\langle 2 \rangle 1$ It is enough to prove (G) pointwise for every pair (M, ξ) .

Because: The Weyl-reduction module proves that the finite linear span of Weyl operators is norm dense in each coarse Weyl algebra. If the state values converge on each generator, linearity gives convergence on that span. Given A and an approximant A_0 in the span with $\|A - A_0\| < \delta$, the state norm-one bound gives

$$|\omega_N(\alpha_M^N(A)) - \rho_M(A)| \leq 2\delta + |\omega_N(\alpha_M^N(A_0)) - \rho_M(A_0)|.$$

Let $N \rightarrow \infty$, then $\delta \downarrow 0$. Positivity, normalization, projective consistency, and continuum identification follow in that module from the same pointwise limits.

$\langle 2 \rangle 2$ Uniformity in ξ on bounded sets is therefore not a hypothesis of the gate.

Because: Step $\langle 2 \rangle 1$ uses only one fixed approximant, hence finitely many fixed Weyl symbols. Quantitative uniform estimates are useful outputs but not logical prerequisites.

⟨2⟩ 3 An N -uniform interacting gap is not an input.

Because: V4 first derives eigenvalue convergence from normal-ordered partition functions and only then obtains $E_1(N) - E_0(N) \rightarrow \gamma_L > 0$. The eventual gap is an output used in the final thermal-to-ground interchange.

⟨2⟩ 4 The right side of (G) denotes the ground state of the specific K_L constructed as the V3 strong semigroup limit.

Because: V3 identifies $H_0 + V_L$ on the cylinder core but does not prove essential self-adjointness of that core restriction. Part II neither needs nor proves uniqueness among other possible realizations. The gate is an identification with the constructed object, in convention $(a, b) = (0, 0)$.

⟨2⟩ 5 This proves ⟨1⟩1. □

2 Why momentum cannot be inserted as a sharp-time Gaussian source

⟨1⟩ 2 Prove the sharp-time momentum-source obstruction.

Let $0 \neq h \in C^\infty(\mathbb{T}_L)$ be real and $C = (-\partial_t^2 - \partial_x^2 + m^2)^{-1}$ on $\mathbb{R}_t \times \mathbb{T}_L$. The formal momentum insertion would use $J = \delta'_0 \otimes h$.

⟨2⟩ 1 The Fourier transform of J is

$$\hat{J}(k_0, k) = ik_0 \hat{h}(k). \quad (2.1)$$

Because: Distributional differentiation in time multiplies the Fourier transform by ik_0 ; the spatial tensor factor supplies $\hat{h}(k)$.

⟨2⟩ 2 Its covariance norm is

$$\|J\|_{H^{-1}}^2 = \frac{1}{2L} \sum_{k \in (\pi/L)\mathbb{Z}} |\hat{h}(k)|^2 \int_{\mathbb{R}} \frac{dk_0}{2\pi} \frac{k_0^2}{k_0^2 + m^2 + k^2}. \quad (2.2)$$

Because: The first-chaos norm of a source is $\langle J, CJ \rangle$; in Fourier variables, C multiplies by $(k_0^2 + m^2 + k^2)^{-1}$. Insert (2.1) and Parseval's factor $1/(2L)$.

⟨2⟩ 3 The right side of (2.2) is infinite.

Because: Choose k_* with $\hat{h}(k_*) \neq 0$. For $|k_0| \geq \sqrt{m^2 + k_*^2}$,

$$\frac{k_0^2}{k_0^2 + m^2 + k_*^2} \geq \frac{1}{2}.$$

The corresponding k_0 integral dominates the integral of $1/2$ over two unbounded rays and therefore diverges. All summands are nonnegative, so one divergent term makes the sum infinite.

⟨2⟩ 4 The same obstruction holds for every spatial lattice.

Because: Only the spatial momentum set and the value of the positive dispersion change. Time remains continuous, and for each nonzero spatial coefficient $k_0^2/(k_0^2 + \gamma_N(k)^2) \rightarrow 1$ as $|k_0| \rightarrow \infty$.

⟨2⟩ 5 Hence $\Phi(\delta'_0 \otimes h)$ is not a Gaussian first-chaos random variable and $e^{i\Pi(h)}$ cannot be represented as $e^{i\Phi(J)}$ with an H^{-1} source.

Because: A Gaussian source functional exists in L^2 precisely when its covariance norm is finite; (2.2) violates that condition.

⟨2⟩ 6 This proves ⟨1⟩2. □

⟨1⟩ 3 Replace the impossible momentum source by the exact jump mean shift.

⟨2⟩ 1 At the free thermal level, compute momentum insertions by the oscillator trace formula.

Because: V1 derives the one-mode expression $\text{Tr}(e^{-tH}D(z))/\text{Tr} e^{-tH} = e^{-|z|^2 \coth(t\gamma/2)/2}$ and takes its finite or trace-dominated infinite product. No path-space derivative source is introduced.

⟨2⟩ 2 At the interacting level, let $e^{i\Pi(p)}$ translate the marked configuration and shift the Gaussian bridge by the unique harmonic path H_p .

Because: V1 solves $-\ddot{H} + \gamma^2 H = 0$, $H(t) - H(0) = \rho$, $\dot{H}(t) = \dot{H}(0)$ and proves the exact Mehler-chain identity. The shifted interaction is the finite Appell sum

$$\mathcal{U}(\varphi + H) - \mathcal{U}(\varphi) = \lambda \int [4H:\varphi^3: + 6H^2:\varphi^2: + 4H^3\varphi + H^4]. \quad (2.3)$$

Thus the momentum variable enters only through a deterministic, bounded harmonic interpolant.

⟨2⟩ 3 This proves ⟨1⟩3. □

3 The executed thermal two-torus route

⟨1⟩ 4 Reduce the gate to eight explicit packages.

Item	Required statement	Executed proof
A1	Continuum K_L , trace class, simple ground, positive gap	V3 Steps ⟨1⟩9–⟨1⟩16
A2	Lattice and continuum marked-slice mean-shift traces	V1 Step ⟨1⟩6; V3 Step ⟨1⟩15
A3	Exact free thermal forms and fixed-coarse rate	V1 Steps ⟨1⟩3, ⟨1⟩7–⟨1⟩8
A4	Same-noise interaction comparison, Wick dictionary, shifted terms	repaired V2 Steps ⟨1⟩1–⟨1⟩8
A5	Uniform plain and shifted exponential integrability	repaired V2 Step ⟨1⟩9; V3 Steps ⟨1⟩5–⟨1⟩6
A6	Fixed- t thermal convergence	V4 Steps ⟨1⟩1–⟨1⟩7
A7	Normal-ordered partition convergence, Laplace spectral transfer, derived gap	V4 Steps ⟨1⟩8–⟨1⟩11
A8	Thermal-to-ground interchange and C^* assembly	V4 Steps ⟨1⟩12–⟨1⟩16

⟨2⟩ 1 A1–A5 imply A6.

Because: A2 writes both thermal functionals on the common A4 probability space. A3 compares deterministic free factors; A4 compares all random exponents in L^2 ; A5 gives uniform integrability; A1 supplies the valid continuum trace identity. V4 performs the three-term quotient subtraction explicitly.

⟨2⟩ 2 A5 and the sourceless part of A6 imply the partition convergence in A7 after subtracting the lattice zero-point energy.

Because: The free normal-ordered product converges by the dispersion estimate, and the interacting factor is $\mathbb{E}e^{-\mathcal{U}_N}$. The unshifted free product with its zero-point factor has no nonzero continuum limit; V4 uses only $\prod_k(1 - e^{-t\gamma_N(k)})^{-1}$.

⟨2⟩ 3 A7 and the A1 continuum gap imply the uniform spectral-tail input for A8.

Because: The exact Laplace-transfer proof gives $E_1(N) - E_0(N) \rightarrow \gamma_L > 0$ and $D_N(t_*) \rightarrow D_\infty(t_*)$. For $t \geq t_*$,

$$D_N(t) \leq e^{-(t-t_*)[E_1(N)-E_0(N)]} D_N(t_*).$$

⟨2⟩ 4 A6 plus this tail input permits the order of limits to be interchanged, and the Weyl criterion then proves (G) on the whole inductive-limit algebra.

Because: V4 Step ⟨1⟩13 writes the three-term triangle inequality, first chooses t uniformly large by the derived gap, and only then sends $N \rightarrow \infty$ using fixed- t convergence. Its final two steps invoke the proved Weyl-reduction module.

⟨2⟩ 5 This proves ⟨1⟩4. □

⟨1⟩ 5 Explain why the slab-diagonal fallback is not load bearing.

⟨2⟩ 1 A slab approximation at time length T has an error controlled by the individual lattice gap $\gamma_{N,L}$.

Because: Spectral expansion gives an excited-state tail with leading factor $e^{-T\gamma_{N,L}}$.

⟨2⟩ 2 Interchanging $T \rightarrow \infty$ and $N \rightarrow \infty$ on that route would require an independent eventual uniform lower bound on $\gamma_{N,L}$.

Because: Without such a bound, no single T makes $e^{-T\gamma_{N,L}}$ small for all sufficiently large N .

⟨2⟩ 3 The thermal route is stronger because the trace partition functions derive precisely that gap through Laplace transfer.

Because: The compact time circle gives trace identities, and convergence of all large-time Laplace transforms determines the ordered atoms of the spectrum. The slab route has no corresponding normalized trace input. It is therefore a fallback, not part of the executed proof chain.

⟨2⟩ 4 This proves ⟨1⟩5. □

4 Scope conclusion

The scoping document contributes one substantive negative result, Step ⟨1⟩2, and the dependency architecture in Step ⟨1⟩4. Its tentative A4 coupling and tentative counterterm were superseded: the executed coupling is V2’s edge-symmetrized same-noise construction, and the equal-point variance offset cancels the tadpole offset, leaving $(a, b) = (0, 0)$. The repaired V2 module further supplies the four-edge term and the fundamental-band/alias partition required for A4 to be true at proof level.